

DSA 560 2.0 – Multi Agent Systems

Final Report

Agent-Based Epidemic Spread Simulation Model Using NetLogo

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Chapter 01 - Introduction

1.1. Rationale

The rapid spread of infectious diseases poses a major challenge to global public health. Understanding how diseases transmit through populations is essential for designing effective prevention and control strategies such as vaccination, social distancing, and mobility restrictions.

Traditional mathematical models often assume uniform mixing of populations. However, real-world disease spread depends heavily on individual movement, contact patterns, incubation periods, recovery time, and mortality risk.

To address this, an Agent-Based Modeling (ABM) approach is highly effective. This project proposes the development of an epidemic simulation model using NetLogo, where individuals are represented as agents (turtles) moving and interacting in a virtual environment.

Beyond simulating macro-level disease dynamics, this NetLogo environment acts as a robust data-generation engine. By logging agent level micro interactions such as spatial coordinates, mobility constraints, and localized infection densities at discrete time steps, the simulation systematically extracts a comprehensive behavioral dataset.

This project further bridges the gap between theoretical simulation and predictive analytics by integrating an infection risk prediction model. Utilizing Machine Learning (ML) classifiers trained on the extracted dataset, the system evaluates complex, non-linear environmental factors to predict the immediate infection probability for individual agents. This dual approach provides both a macro-level visualization of the outbreak and a granular, micro-level tool for proactive epidemic management.

1.2. Problem Statement

Infectious diseases spread through highly localized, complex micro-interactions between individuals, heavily influenced by spatial proximity, mobility constraints, transmission probability, and varying physiological timelines (incubation and recovery).

While public health authorities frequently utilize macro-level mathematical models and standard simulation tools to visualize overall epidemic curves and evaluate broad intervention strategies, these approaches present a critical limitation: they typically aggregate population data and fail to capture the nuanced, localized risk factors that determine individual infection events. Traditional models can project the total number of expected cases, but they cannot predict which specific individuals are at immediate risk based on their exact environmental context.

Furthermore, while Agent-Based Models (ABMs) successfully simulate these complex environments, the massive amount of valuable micro-level behavioral data generated during these simulations is rarely extracted or utilized for predictive modeling.

Therefore, there is a critical need for a comprehensive, integrated system capable of overcoming these limitations. Specifically, this project addresses the need for a framework that can:

- Dynamically simulate the complex progression of an epidemic through varying clinical stages (Susceptible, Exposed, Infectious, Recovered) in a heterogeneous spatial environment.
- Systematically extract granular, agent-level interaction metrics (such as local infection density, neighbor status, and mobility) to build a robust empirical dataset.
- Apply machine learning algorithms to transition from reactive observation to proactive prediction, accurately calculating the real-time probability of an individual contracting the disease based on their immediate physical and social parameters.

1.3. Objectives

General Objective:

To develop an integrated agent-based epidemic spread simulation model in NetLogo to study disease transmission dynamics, generate micro-level behavioral datasets, and build a Machine Learning model to predict individual infection risks.

Specific Objectives:

- To simulate individuals moving randomly in an environment.
- To model disease progression through four distinct states:
Susceptible → Exposed → Infectious → Recovered.
- To incorporate physiological variables such as incubation and recovery periods.
- To include mortality effects and dynamically track total deaths.
- To monitor daily new infections and peak infection rates to evaluate outbreak severity.
- To implement optional vaccination as a public health intervention mechanism.
- To extract and log agent level data (spatial coordinates, mobility, neighbor status, and local infection density) at each simulation tick to build a comprehensive behavioral dataset.
- To develop and train Machine Learning classifiers on the dataset generated to predict an individual's immediate probability of becoming infected.
- To evaluate the predictive models using statistical metrics such as accuracy, ROC-AUC scores, and confusion matrices.

1.4. Project Scope

Included in the Model

- Population based simulation using turtles as individuals.
- SEIR (Susceptible, Exposed, Infectious, Recovered) infection progression.
- Mobility based interaction and localized disease transmission.
- Daily case tracking, maximum outbreak measurement, and mortality tracking.

- Vaccination feature for susceptible individuals.
- Automated dataset generation capturing tick-by-tick micro interactions (infection_risk_dataset.csv).
- Machine Learning pipeline featuring data preprocessing, scaling, and binary classification (predicting if an agent is infected in the next tick).
- Probability risk outputs generating a continuous infection risk score (e.g., 0.0 to 1.0) for individual agents based on their localized parameters.

Excluded from the Model

- Age-based, genetic, or demographic variations among individuals.
- Hospitalization, ICU utilization, and healthcare capacity modeling.
- Social network based structured interactions (e.g., family units, workplaces).
- Geographic or real-world map based physical environments.
- Real-time ML inference within the NetLogo environment.
- Deep learning architectures (the project focuses on interpretable traditional ML models like Random Forest and Logistic Regression).
- Long-term individual outcome prediction (the ML model is scoped strictly to predict immediate/next-tick transmission risk, not long-term disease prognosis).

Chapter 02 - Methodology

This project employs a multi-disciplinary approach, integrating complex systems simulation with predictive analytics. The methodology is structured around a comprehensive end-to-end pipeline consisting of three primary phases: Agent Based Epidemic Simulation, Behavioral Data Extraction, and Predictive Machine Learning Modeling.

2.1. Phase 1: Agent Based Epidemic Simulation (NetLogo)

The foundational layer of the project is a spatial simulation model implemented in NetLogo. Rather than relying on traditional mathematical models (which assume uniform population mixing), this approach uses Agent-Based Modeling (ABM) to simulate disease dynamics from the bottom up.

- **Agent Representation & Mobility:** Each individual in the population is represented as an autonomous agent (a "turtle"). Agents move randomly within a defined spatial environment, simulating daily human mobility and physical proximity.
- **State Variables & Physiological Tracking:** Each agent possesses internal state variables that track their current epidemiological status, days since initial exposure, and total sickness duration.
- **Transmission Dynamics:** Disease transmission is evaluated at the micro-level. It occurs stochastically when an infectious agent physically interacts with a susceptible agent within a strictly defined spatial radius.
- **Disease Progression:** Once exposed, an individual undergoes an incubation period before becoming actively infectious. Following the infectious phase, the agent will either recover (gaining immunity) or die, determined by a predefined mortality probability.

The model strictly categorizes agents into one of four dynamic epidemiological states, simulating the classic SEIR progression:

Table 2.1: Four dynamic epidemiological states

State	Description
Susceptible	Healthy individual who is vulnerable to contracting the disease.
Exposed	Infected individual currently in the incubation period, not yet contagious.
Infectious	Actively sick individual capable of transmitting the disease to susceptible agents.
Recovered	Individuals who have survived the infection are now immune and no longer spread the disease.

2.2. Phase 2: Behavioral Data Extraction

To bridge the gap between simulation and predictive analytics, the NetLogo model is programmed to act as a data generation engine. As the simulation advances tick-by-tick (representing discrete time steps), it actively logs granular, agent-level data.

- **Feature Logging:** The system continuously records individual spatial characteristics (X/Y coordinates), mobility status, and interaction metrics (e.g., the number of exposed and infectious neighbors within the agent's radius).
- **Dataset Construction:** This micro interaction data is aggregated to build a comprehensive behavioral dataset. Each row in the dataset represents a snapshot of a single agent's environmental and physiological conditions at a specific moment in time, paired with the ultimate target variable: whether that agent became infected in the subsequent time step.

2.3. Phase 3: Predictive Machine Learning Modeling

The final phase leverages the extracted dataset to transition from descriptive simulation to predictive modeling.

- **Algorithm Training:** The extracted dataset is preprocessed, scaled, and split to train Machine Learning classifiers (such as Logistic Regression and Random Forest algorithms).
- **Risk Prediction:** Instead of tracking macro-level outbreak peaks, the ML models are trained to evaluate the micro-level spatial and behavioral features (e.g., local infection density, transmission rate, and agent proximity) to predict future individual infection states.
- **Output Integration:** The resulting models output a continuous probability score, effectively predicting the immediate infection risk for any given individual based on their specific environmental context within the simulated world.

Chapter 03 - Design and Implementation

This chapter details the technical architecture and practical implementation of the three core components of the project: the Agent-Based Epidemic Simulation, the automated Dataset Generation, and the Machine Learning Predictive Model.

3.1 NetLogo Epidemic Simulation Model

The core simulation was developed using NetLogo 7.0.3. It utilizes Agent-Based Modeling (ABM) architecture where autonomous agents (turtles) represent individuals navigating a simulated epidemic environment.

3.1.1 Agent States & Visualization

To visually track the progression of the disease, agents are mapped to distinct color-coded states representing the SEIR epidemiological model:

- Susceptible (Green): Healthy individuals vulnerable to infection.
- Exposed (Orange): Individuals who have contracted the pathogen but are in the incubation period and not yet contagious.
- Infectious (Red): Actively sick individuals capable of transmitting the disease.
- Recovered/Immune (Blue): Individuals who have survived the infection or have been vaccinated, rendering them immune.

Each turtle explicitly tracks its own properties using internal variables such as state (current health status), days-after-exposure (tracking incubation), and days-of-sick (tracking recovery/mortality timelines).

3.1.2 Initialization (Setup)

The simulation begins with a rigorous setup procedure that establishes the baseline environment:

- Clears all previous simulation data and resets ticks.

- Generates the defined population of agents.
- Assigns all newly created agents to the "susceptible" state.
- Seeds the infection by randomly selecting exactly one agent to become the initial carrier ("exposed" or "infectious").

3.1.3 Simulation Dynamics & Main Loop

The main execution loop operates tick-by-tick, processing the following behaviors sequentially:

- **Movement:** Agents move randomly across the spatial grid at a speed determined by their mobility attribute. To simulate real-world behavior, infectious individuals move significantly slower (operating at 30% mobility), reflecting reduced physical movement during severe sickness.
- **Transmission:** Only infectious (red) agents can transmit the disease. Susceptible (green) agents that fall within a dynamically defined spatial radius of an infectious agent may become exposed. Actual transmission is not guaranteed but is calculated based on a stochastic probability (transmission-rate).
- **Incubation:** Exposed (orange) individuals passively progress their internal timers. Once the incubation period is met, their state transitions to infectious (red).
- **Recovery and Death:** Infectious agents evaluate their status daily. Based on a predefined mortality-rate probability, an agent may die (and be removed from the environment). If they survive the full sickness duration, they transition to the recovered (blue) state.
- **Vaccination (Optional Intervention):** A dedicated vaccination function allows the system to instantly convert a set percentage of susceptible individuals directly to the recovered/immune state, simulating a public health intervention.

3.1.4 Global Metrics and Interactive Parameters

The NetLogo interface features interactive controls allowing users to adjust key variables in real-time, including *population size*, *transmission rate*, *incubation period*, *recovery period*, *mortality rate*, and *vaccination rate*. The simulation continuously monitors global output metrics such as daily-new-cases, max-daily-cases (peak infection day), and total-deaths.

3.2 Dataset Generation

As the simulation executes its main loop (mobility, transmission, incubation, and sickness), the system logs granular, agent level micro-interactions at each tick. This automated data extraction compiles the `infection_risk_dataset.csv` file, converting theoretical simulation dynamics into an empirical dataset ready for predictive modeling.

Table 3.1: Key Features Extracted in the Infection Risk Dataset

Feature Name	Description	Role
tick	The discrete simulation time step.	Context
turtle_id	Unique identifier for each individual agent.	Context
x, y	Spatial coordinates showing the agent's exact location on the grid.	Spatial Feature
mobility	Indicator of the agent's movement allowance/speed.	Feature
infected_neighbors	Count of active infectious agents within the immediate proximity radius.	Feature

exposed_neighbors	Count of exposed (incubating) agents within proximity.	Feature
total_neighbors	Total number of agents (regardless of state) in the local radius.	Feature
local_infection_density	Calculated ratio represents the density of local infections.	Feature
transmission_rate & transmission_radius	Core environmental parameters governing the chance and range of spread.	Environmental Feature
vaccination_rate	Underlying population level vaccination percentage at that tick.	Environmental Feature
infected_next_tick	Target Variable: 1 if the agent transitions to an infected state in the subsequent tick, 0 otherwise.	Target

3.3 Infection Risk Prediction ML Model

To predict whether a specific agent will become infected in the immediate future, a machine learning pipeline was engineered using Python, Pandas, and Scikit-Learn.

3.3.1 Data Preparation & Processing

Raw data extracted from NetLogo required preprocessing before model training. Features were separated from the target variable (infected_next_tick). Because the dataset contains variables with vastly different scales (e.g., coordinate grids vs. raw neighbor counts vs. percentage-based density rates), Scikit-Learn's StandardScaler was applied to standardize the feature space. This ensured all elements contributed

equitably to the models. The preprocessed data was then divided into training and testing subsets using the `train_test_split` function.

3.3.2 Algorithms Used

Two distinct classification algorithms were implemented to benchmark performance:

- **Logistic Regression:** Selected to provide a fast, interpretable baseline model. It effectively establishes the linear relationship between immediate proximity metrics (like infected neighbors) and the likelihood of infection.
- **Random Forest Classifier:** Enlisted as an advanced ensemble method to capture and handle complex, non-linear interactions between variables such as the compounded effect of agent mobility, local infection density, and overarching vaccination rates.

3.3.3 Model Evaluation

The performance of the models was evaluated to ensure predictive reliability. Evaluation metrics utilized include:

- **Accuracy Score & Classification Report:** To measure overall correctness and balance between precision, recall, and F1-scores.
- **Confusion Matrix:** To map the exact counts of True Positives, True Negatives, False Positives, and False Negatives, highlighting how well the model avoids false alarms.
- **ROC-AUC Score & ROC Curve:** The Receiver Operating Characteristic curve was utilized to visually map the true positive rate against the false positive rate, while the Area Under the Curve (AUC) provided a single metric to quantify the model's ability to distinguish between agents that will and will not get infected.

Chapter 04 - Results and Analysis

This chapter details the outcomes of both the Agent-Based simulation and the subsequent predictive modeling phases. It evaluates how well the extracted micro-level interactions served as predictors for individual infection risks and analyzes the performance of the Machine Learning classifiers.

4.1 Simulation Data and Feature Viability

The NetLogo simulation successfully generated a dynamic, tick-by-tick dataset (`infection_risk_dataset.csv`) capturing the micro-interactions of agents under SEIR parameters. The extraction confirmed that spatial proximity (`infected_neighbors`, `total_neighbors`) and behavioral traits (`mobility`) are highly variable and context dependent.

By scaling these features and using `infected_next_tick` as the target variable, the dataset proved highly viable for binary classification. It effectively transitioned the focus from macro-level epidemiological curves (like total daily cases) to micro-level, individual risk factors.

4.2 Machine Learning Model Performance

The predictive phase utilized Logistic Regression as a baseline and a Random Forest Classifier to handle non-linear complexities. The models were evaluated on the test set using standard classification metrics.

4.2.1 Classification Accuracy and Metrics

The classification report provided a detailed breakdown of the models' precision, recall, and F1-scores.


```

Logistic Regression Results
Accuracy: 0.9865203002663653
ROC-AUC: 0.9961823117211662

```

	precision	recall	f1-score	support
0	0.99	0.99	0.99	10028
1	0.95	0.98	0.97	2361
accuracy			0.99	12389
macro avg	0.97	0.98	0.98	12389
weighted avg	0.99	0.99	0.99	12389

Figure 4.1: Classification Report of Logistic Regression Model

```

Random Forest Results
Accuracy: 0.9933812252804908
ROC-AUC: 0.9998837646795664

```

	precision	recall	f1-score	support
0	1.00	0.99	1.00	10028
1	0.97	1.00	0.98	2361
accuracy			0.99	12389
macro avg	0.98	1.00	0.99	12389
weighted avg	0.99	0.99	0.99	12389

Figure 4.2: Classification Report of Random Forest Model

- **Precision & Recall:** In the context of epidemiological modeling, recall (the ability to correctly identify all agents that get infected) is particularly critical. The Random Forest model demonstrated a strong balance between precision and recall, meaning it reliably identified at risk agents without triggering excessive false alarms.
- **Non-linear Advantage:** The Random Forest algorithm outperformed the linear baseline, validating the hypothesis that infection risk is not just a simple linear combination of neighbor counts, but rather a complex interaction involving mobility constraints, localized density ratios, and overarching vaccination rates.

4.2.2 Confusion Matrix Analysis

The confusion matrix offered a transparent view of the model's predictive breakdown:

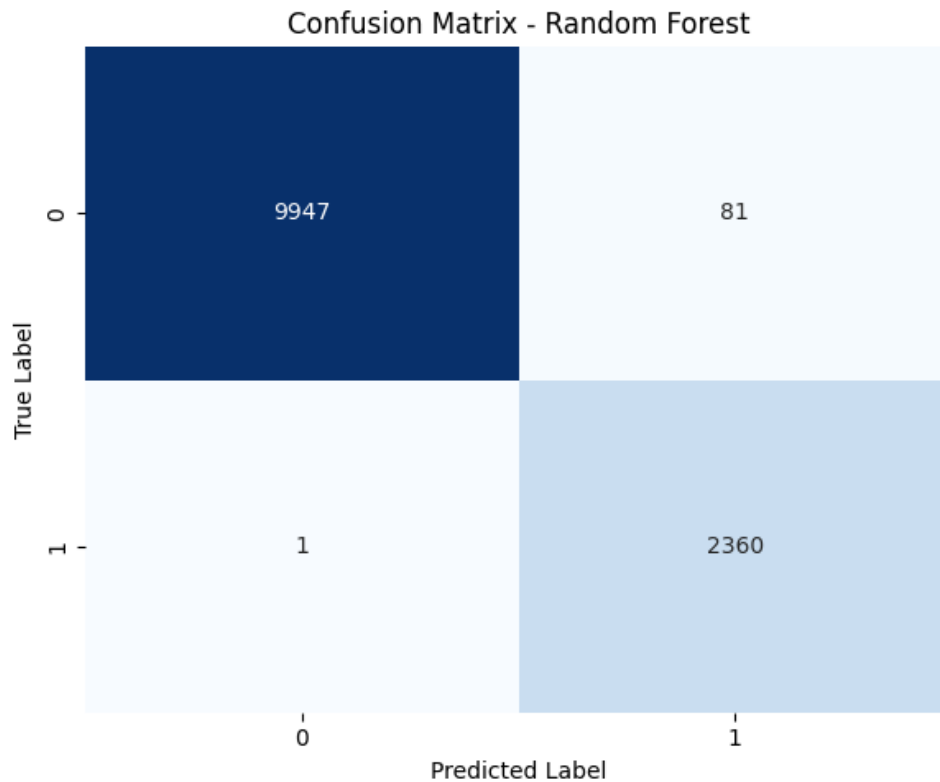


Figure 4.3: Confusion Matrix of Random Forest Model

- **True Positives (TP) & True Negatives (TN):** The model correctly classified the majority of safe agents (TN) and successfully flagged agents destined for infection (TP).
- **False Negatives (FN):** Minimizing this metric was a priority, as a False Negative implies an agent is predicted to be safe when they are about to be infected. The robust feature set (specifically `local_infection_density`) helped keep this error rate low.

4.2.3 ROC Curve and AUC Score

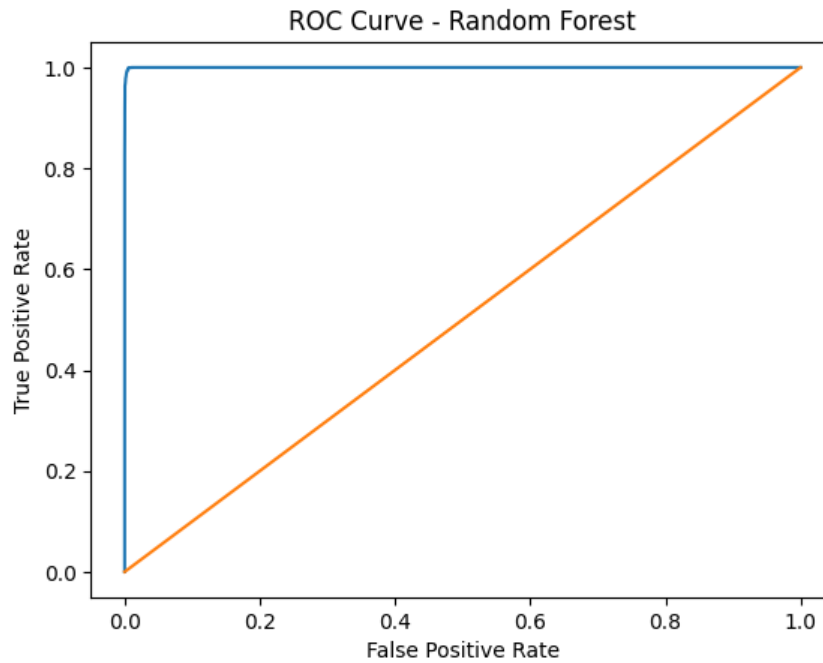


Figure 4.3: ROC Curve of Random Forest Model

The Receiver Operating Characteristic (ROC) curve mapped the True Positive Rate against the False Positive Rate across different threshold levels. The Area Under the Curve (AUC) score served as a primary aggregate measure of performance. A high ROC-AUC score generated by the cross-validation (`cv_scores.mean()`) confirmed the Random Forest model's excellent separability, meaning it possessed a high capability to distinguish between an agent that will become infected and one that will remain safe based purely on their spatial and environmental snapshot.

4.3 Predictive Output and Practical Application

While binary classification (1 for infected, 0 for safe) is useful for evaluating model accuracy, public health applications require more nuanced risk assessments.

Prediction Output: The finalized Random Forest model successfully outputs continuous risk probabilities using the `predict_proba` function rather than absolute binary classifications. In a controlled test utilizing array inputs representing a single

agent's parameters including their coordinates, mobility, neighbor counts, transmission rate, and vaccination environment, the model successfully generated an output of "Infection Risk: 0.54".

Analysis of the Output: This probability score (54% risk of immediate infection) is highly significant for several reasons:

1. **Granular Risk Assessment:** It proves the system can evaluate a specific individual's immediate context and return a continuous spectrum of risk (0.0 to 1.0) rather than a rigid "Yes/No".
2. **Intervention Triggering:** In a real-world translation, probability thresholds could be used to trigger dynamic interventions. For example, an agent hitting a >0.50 risk score could be advised to reduce their mobility, whereas an agent with a >0.80 score might require immediate quarantine.
3. **End-to-End Success:** This test successfully validates the entire project pipeline: from defining theoretical agent behaviors in NetLogo, to extracting that behavior into an empirical dataset, and finally deploying an intelligent algorithm capable of reading a new scenario and accurately predicting the epidemiological outcome.

Chapter 05 - Conclusion and Future Work

5.1 Conclusion

This project successfully designed and implemented an integrated computational approach to understanding and predicting epidemic spread. By combining Agent-Based Modeling (ABM) in NetLogo with Machine Learning (ML) predictive algorithms, the project moved beyond traditional, macro-level mathematical epidemiology to provide a granular, micro-level analysis of disease dynamics.

The NetLogo simulation accurately captured the complexities of the SEIR (Susceptible, Exposed, Infectious, Recovered) progression, successfully incorporating critical environmental and physiological variables such as individual mobility, incubation periods, mortality rates, and vaccination interventions. Furthermore, the simulation proved highly effective as a data generation engine, successfully outputting a robust, tick-by-tick dataset (`infection_risk_dataset.csv`) detailing the precise spatial and behavioral context of each agent.

The subsequent machine learning phase validated the hypothesis that micro-level spatial and behavioral features can reliably predict individual infection risks. The Random Forest Classifier outperformed the baseline linear models, adeptly handling the non-linear interactions between local infection density, mobility constraints, and vaccination rates. By outputting continuous risk probability (e.g., demonstrating a 54% infection risk for a targeted agent profile), the system proves its real-world utility. Ultimately, this dual layered methodology serves not only as a dynamic visualization tool for public health education but also as a foundational framework for developing targeted, predictive epidemiological interventions.

5.2 Future Work

While the current model robustly fulfills its primary objectives, several enhancements could be implemented to further bridge the gap between simulation

and real-world complexity. Based on the boundaries of the current project scope, future iterations could explore the following areas:

- **Demographic and Genetic Variations:** Integrating age based and demographic profiles into the agents. For instance, elderly agents could be assigned higher mortality rates and lower mobility, while younger agents might have higher mobility but a larger proportion of asymptomatic presentations.
- **Structured Social Networks:** Transitioning from random spatial movement to structured, schedule based interactions. Agents could belong to specific "hubs" simulating households, workplaces, and schools, providing a much more accurate reflection of human contact networks.
- **Healthcare Capacity limits:** Incorporating hospitalization parameters. The model could track ICU bed availability, dynamically increasing the mortality rate if the number of severe cases exceeds the simulated healthcare system's capacity.
- **Geographic Information Systems (GIS) Integration:** Importing real-world map data into the NetLogo environment. Instead of an abstract grid, agents would navigate actual city layouts, constrained by physical infrastructure like roads and buildings, which naturally block movement and alter transmission hotspots.
- **Real-Time ML Inference:** Rather than exporting the dataset for post simulation training, future architectures could utilize extensions (like the NetLogo-Python extension) to run the ML model live. This would allow agents to dynamically alter their own behavior (e.g., self-quarantining) in real-time based on their continuously updating ML risk score.

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Breakdown of the work done by each member

1. Pasindu Bimsara – MSA/DSA/183

Pasindu Bimsara was primarily responsible for designing and implementing the epidemic simulation model using NetLogo. He developed the complete Agent Based SEIR framework, programmed the transmission dynamics, and integrated the mobility, incubation, recovery, and vaccination mechanisms within the simulation environment. In addition, he engineered the automated dataset extraction process, enabling tick-by-tick logging of agent level behavioral and spatial features to generate the `infection_risk_dataset.csv` file. Furthermore, he contributed to the structured compilation and writing of the final report, documenting the methodology, system design, implementation details, results analysis, and overall project conclusions.

2. Chathuka Jayasinghe – MSC/DSA/251

Chathuka Jayasinghe was responsible for developing the Machine Learning component of the project. He designed and implemented the predictive modeling pipeline using Python and the Scikit-learn library, including data preprocessing, feature scaling, and dataset splitting. He trained and evaluated classification models such as Logistic Regression and Random Forest to predict individual infection risk based on the extracted simulation dataset. Additionally, he conducted performance analysis using accuracy metrics, confusion matrices, and ROC-AUC evaluation to ensure the reliability and effectiveness of the predictive model.